Combinations with Repetition - Cookies

- Suppose that a cookie shop has n different kinds of cookies. In how many different ways can r cookies be chosen? There are no constraints on how many cookies of each kind are chosen (e.g., all may be of the same kind)
- **Solution**
  Equivalently, I wish to distribute r indistinguishable cookies to n distinguishable cookie boxes.
- I indicate cookies with r stars. Instead of n boxes, I use $n-1$ bars to indicate their boundaries.
- A selection corresponds to a bars and stars arrangement
- Example: **|*|*|** (2 cookies from first and fourth kind, 1 from the rest).
- # of arrangements = # combinations with repetition =
- = # strings of length $(r+n-1)$ by reordering r stars and $n-1$ bars =
- # select positions of r stars and among possible $(r+n-1)$ positions
- $C(r+n-1,r)=(r+n-1)!/r!(n-1)!$.